The Geometry of Gravitational-Wave Inference: reconstructing component-mass posterior orientation from the chirp-mass curve

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Abstract

The parameter-estimation posteriors of compact-binary gravitational-wave events have a shape that is usually treated as a nuisance. We show instead that this shape is highly structured: the principal-axis orientation of each source-frame (m_1, m_2) posterior can be *reconstructed* from two one-dimensional summaries of that same posterior — its median chirp mass and its mass-ratio marginal — using the constant-chirp-mass curve and *no coefficients calibrated on the validation catalogs*. This is a posterior-compression result, not a prediction from source medians alone: the q marginal is an input taken from the posterior being reconstructed, and we quantify how much of the accuracy it supplies. Derived on GWTC-3 and locked before opening later data, the reconstruction reaches a median mismatch of 1.26° (GWTC-4.0/O4a) and 1.22° (GWTC-5.0/O4b) on elongated events, against 5–7° for the local tangent approximation that underlies rapid parameter-estimation tools. The result is not the trivial statement that a curve fits a curved posterior: substituting any other event's mass-ratio marginal degrades it by 2.5–12×, and the achieved error lies below the *minimum* of 300 catalog-stratified permutations in every catalog. It also transfers across waveform families — inputs from one family recover a different family's separately inferred axis to 2.1°, against a 2.0° reference scale set by how much those families disagree with each other — so it is not an artifact of reusing one posterior's sampling noise. The residual $\sim$1° is *not* sampling noise but a real systematic, roughly six times the Monte Carlo resolution of the released samples; finite posterior thickness, which the zero-thickness curve model omits by construction, is supported out-of-sample as a contributor, while arc-varying thickness is not established. We report the result as a reconstruction law and a diagnostic, and explicitly not as a test of general relativity: the posteriors were produced with general-relativistic waveform models, so their internal geometry cannot bound deviations from it.

1 Introduction

Gravitational-wave (GW) parameter estimation delivers, for each detected compact-binary coalescence, a posterior probability distribution over the source parameters. It is a commonplace that the chirp mass $\mathcal{M}_c = (m_1 m_2)^{3/5}/(m_1 + m_2)^{1/5}$ is the best-measured mass combination and that the component-mass posterior forms an elongated "banana" lying along near-constant-$\mathcal{M}_c$ contours [1; 2; 4]. This paper asks a sharper question: is the *geometry* of that posterior — specifically the orientation of its principal axis — quantitatively *reconstructible*, event by event, from a small number of one-dimensional summaries with no coefficients calibrated on the validation data, and does the reconstruction survive out-of-sample on later, disjoint event catalogs?

We answer yes. The organizing thesis is that the uncertainty geometry of GW inference is *measurement optics*: it is a property of what the instrument and signal can and cannot resolve, largely independent of source astrophysics. This viewpoint is not new mathematically — it is the content of the "sloppy model" framework [7] and of principal curves [8]. Our contribution is to make it quantitative and validated out-of-sample for GW posteriors, and to use it as a systematics diagnostic. We do *not* present it as a test of general relativity; Sec. 7 explains why the geometry of GR-generated posteriors cannot serve that purpose.

2 The curved chirp-mass law

2.1 Definition

For an event with source-frame samples $\{(m_1, m_2)\}$, let $\psi_{\rm meas}$ be the position angle of the principal axis of the sample covariance, and let the axis ratio axr be the ratio of its principal standard deviations. The constant-$\mathcal{M}_c$ *curve* through the plane is

$$m_1(q) = \mathcal{M}_c\,(1+q)^{1/5} q^{-3/5}, \qquad m_2 = q\,m_1, \tag{1}$$

where $q = m_2/m_1 \in (0,1]$ and $\mathcal{M}_c$ is the event's median source-frame chirp mass. The predicted orientation $\psi_{\rm curve}$ is the principal axis of this curve evaluated over the event's own q marginal. The local linear approximation $\psi_{\rm tangent}$ is the tangent to the contour at the median point. No coefficient is calibrated on the validation catalogs: every ingredient is supplied by the event itself.

It is important to be precise about what this is. The q marginal is an input drawn from the very posterior whose orientation is being reconstructed, so this is a *compression* statement — a two-dimensional posterior's orientation recovered from one scalar and one one-dimensional marginal, using no information from the (m_1, m_2) covariance — and not a prediction of posterior shape from source medians or from physics external to the posterior. Sec. 2.3 quantifies how much of the accuracy that marginal supplies.

2.2 Out-of-sample validation across three catalogs

The law was derived from GWTC-1/2.1/3 events and its prediction placed on record before any later-catalog file was opened. Table 1 reports the locked decision on two subsequent, disjoint catalogs. On elongated events (axr ≥ 3, where orientation is well defined) the median mismatch $|\psi_{\rm curve} - \psi_{\rm meas}|$ is 1.26° (GWTC-4.0/O4a) and 1.22° (GWTC-5.0/O4b); the curve beats the tangent on the full catalogs, and the tangent-law replication statistics reproduce the training-set values. The catalogs are mutually disjoint (zero shared events).

Figure 1 shows the per-event comparison behind the table: every elongated event is drawn as a line from its tangent error to its curve error, so the reader sees the whole distribution rather than a summary.

2.3 The reconstruction is not trivial

A curve fitting a curved posterior better than a straight line is unsurprising. The substantive question is whether the *event's own* q marginal matters, or whether any plausible marginal would do. It matters (Fig. 2). Replacing it with a marginal pooled over the catalog, or with another event's, degrades the median error by 2.5–12×; and the achieved error lies below the *minimum* of 300 catalog-stratified permutations in each catalog, so the permutation p-value is $< 1/300$ throughout.

	GWTC-3 (train)	GWTC-4.0/O4a	GWTC-5.0/O4b
median $\lvert\psi_{\text{curve}} - \psi_{\text{meas}}\rvert$ (elongated)	—	**1.26°**	**1.22°**
median $\lvert\psi_{\text{curve}} - \psi_{\text{meas}}\rvert$ (all)	—	7.53°	5.41°
median $\lvert\psi_{\text{tangent}} - \psi_{\text{meas}}\rvert$ (all)	7.49°	9.36°	8.23°
chirp-vs-total win fraction	81%	86%	88%
Spearman($\lvert\Delta\psi_{\text{tangent}}\rvert$, axr)	−0.42	−0.40	−0.69
N events (elongated)	—	86 (19)	104 (32)

Table 1: **Locked scores, computed on the full released posterior samples.** These are the preregistered out-of-sample numbers and are quoted throughout the abstract and text. They are *not* identical to the values in Figures 1–2, which are recomputed from a fixed resampled extract of the same posteriors (1.32° and 1.00° for O4a and O4b) so that every downstream analysis shares one deterministic input; see Sec. 5.1. The two differ by up to 0.2°, which is resampling noise, not disagreement, and we report both rather than silently reconciling them. The curve lands within $\sim 1.2°$ of the measured orientation on elongated events in two later, disjoint event catalogs. Note the sign of the last row: the tangent's error *decreases* as elongation grows (Spearman ≈ -0.4 to -0.7), i.e. orientation becomes better defined for more elongated posteriors. An earlier draft of this work described the residual as growing with elongation; that was wrong. The distinct quantity that does grow is arc length (the q range), and its effect on the tangent error is weak and inconsistent out of sample ($\rho = +0.26, +0.02, +0.12$), so we do not rest the argument on it.

We report the permutation distribution rather than a single shuffled assignment: one draw is not a null, and re-drawing it moved the O4a figure from 3.04° to 8.62°.

A sharper check addresses the circularity concern directly. Using waveform family A's median chirp mass and q marginal to reconstruct family B's *separately inferred* principal axis — never touching B's own geometry — gives 2.25° (A→B) and 2.93° (B→A) over 64 events elongated in both, against a 2.03° reference scale set by how far the two families' measured axes sit from each other, and against tangent baselines of 3.50° and 6.20° in the same two directions. The reconstruction therefore transfers between separately inferred posteriors of the same event and is not an artifact of reusing one posterior's Monte Carlo noise. These are *not* independent inferences, however: the two waveform families analyse the same strain with shared calibration and priors, so the test excludes reuse of one posterior's sampling noise, not correlated modelling error common to both. We call 2.03° a reference scale, not an error floor: a reconstruction could in principle denoise a posterior functional and beat the raw A–B disagreement.

3 Why the geometry is structured

The structure is expected. The inspiral phase is dominated at leading (0PN) order by $\mathcal{M}_c$, so there is one sharply-measured ("stiff") direction — constant $\mathcal{M}_c$ — and a poorly-measured ("sloppy") direction transverse to it [7], and the posterior's long axis therefore lies along the constant-$\mathcal{M}_c$ level set. We use that language for the concept only. The stronger "hyperribbon" picture, in which the eigenvalue spectrum spans many orders of magnitude, does *not* follow from what we measure: in the two-dimensional mass plane the covariance eigenvalue ratio has a median of 3.3, with 29% of events exceeding one order of magnitude and 3% exceeding two. Sloppiness in that sense is a property of the full parameter space, which a two-dimensional marginal cannot establish.

There is a sharper way to say what the tangent term is. The principal axis of a Gaussian is the leading eigenvector of its covariance, which for a likelihood expanded quadratically about its

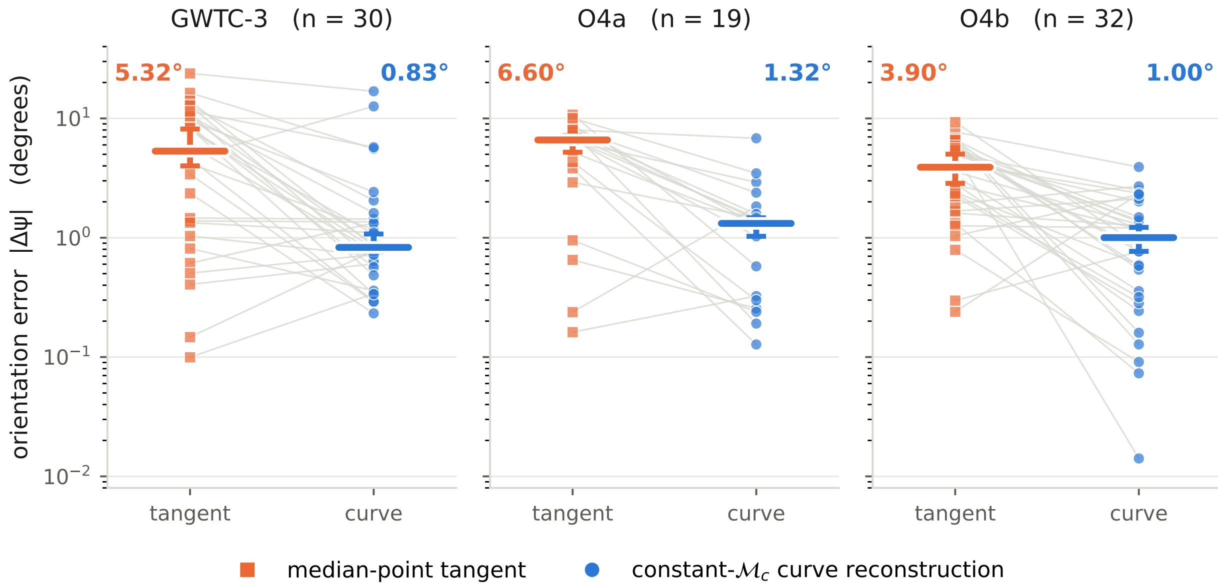

Figure 1: Reconstruction versus the local linear approximation, on elongated events (axr ≥ 3) in three disjoint event catalogs. Each thin line is one event, joining its median-point tangent error to its constant-$\mathcal{M}_c$ curve reconstruction error; heavy bars are catalog medians with a bootstrap interval computed over events. Medians: GWTC-3 ($n = 30$): $5.32° \to 0.83°$, O4a ($n = 19$): $6.60° \to 1.32°$, O4b ($n = 32$): $3.90° \to 1.00°$. The vertical scale is logarithmic because per-event errors span roughly 0.01–24°. The interval is a bootstrap over the event sample, not the per-event Monte Carlo resolution of Sec. 4.

peak is the local tangent to the degeneracy; the tangent error therefore measures exactly the error incurred by treating the posterior as Gaussian. (We claim only this elementary geometric fact. The asymptotic statement that posteriors approach a Gaussian with inverse-Fisher covariance is standard but is not what the argument needs, and we do not lean on it.) On elongated events that error is 5.02°, against 1.03° once the arc is accounted for: the Gaussian limit is wrong about orientation by roughly a factor of 5 at realistic signal-to-noise. That gap — not the asymptotic statement — is what this paper measures and corrects. The distinction between the tangent and the curve is the distinction between linear principal-component analysis and a *principal curve* [8]: the principal axis of samples spread along a curved arc is a chord, which is rotated off the tangent at the median point.

We can do better than invoke that framework by name. Hastie & Stuetzle [8] define a principal curve by *self-consistency*: $f(t) = \mathbb{E}[X \mid t_f(X) = t]$, i.e. every point of the curve is the mean of the data projecting onto it. This is directly testable for the constant-$\mathcal{M}_c$ curve, and it fails: projecting each posterior sample to its nearest curve point and comparing that point to the mean of what lands there leaves a perpendicular offset of about 3.1% of the posterior's own scale. The constant-$\mathcal{M}_c$ curve is therefore *not* a principal curve of the posterior — and the size of that failure is what predicts the reconstruction residual (Sec. 4).

We deliberately do not dress this in stronger language than the analysis supports. In particular we do not invoke Čencov's uniqueness theorem for the Fisher–Rao metric: our measured object — the principal-axis angle of a sample covariance in fixed (m_1, m_2) coordinates — is demonstrably *not* invariant (Sec. 5 reports 1.03°, 0.31° and 0.77° in source-frame, detector-frame and log-mass

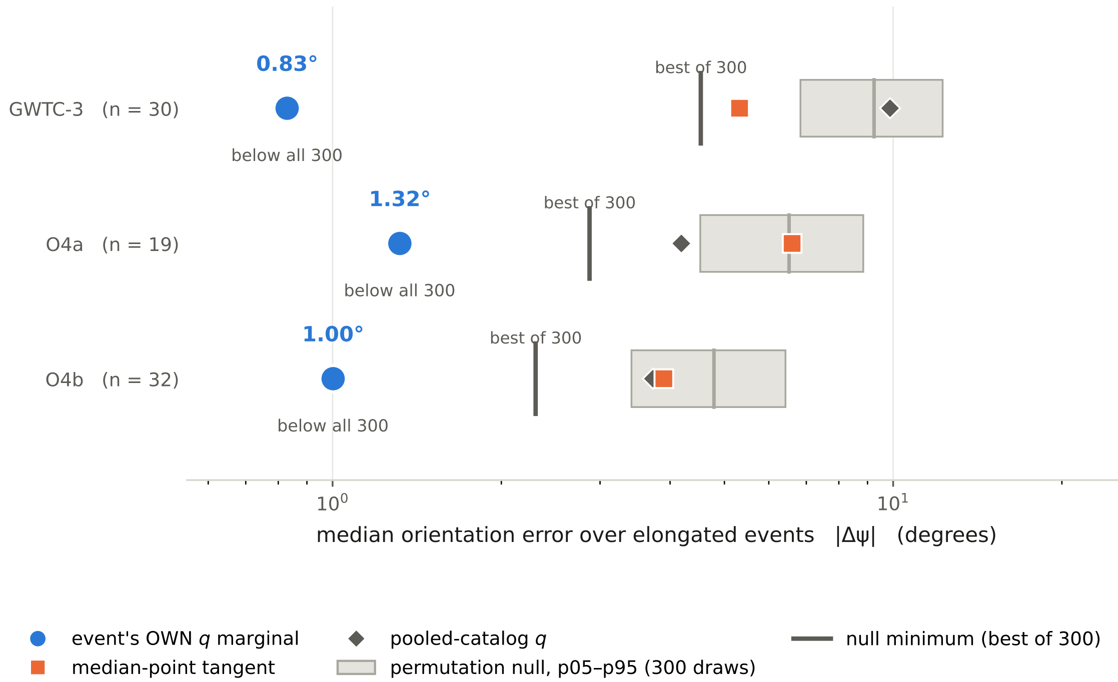

Figure 2: The reconstruction requires the *event's own* mass-ratio marginal. For each catalog the blue circle is the median error using the event's own q marginal; the orange square is the median-point tangent; the grey diamond substitutes a q marginal pooled over the catalog. The grey band is a **permutation null**—the 5th–95th percentile of the catalog-median error when each event is given a *different* event's q marginal, over 300 catalog-stratified draws—and the dark tick marks that null's *minimum*, i.e. the single best of those draws. Own-q lies below the null minimum in every catalog (GWTC-3: own-q 0.83° vs. null minimum 4.54°; O4a: own-q 1.32° vs. null minimum 2.87°; O4b: own-q 1.00° vs. null minimum 2.30°), so the permutation p-value is $< 1/300$ in each. **The band is a null distribution: it is not a confidence interval, not coverage, and not an uncertainty on own-q.**

coordinates), so appealing to an invariance theorem would assert the opposite of what we measured. We compute no Fisher–Rao quantity, and the angle it changes under a change of mass coordinates, and we state the convention rather than implying invariance. Sec. 5 reports the frame and coordinate dependence explicitly. The closest published neighbours stop short of this measurement in identifiable ways. Hannam et al. [4] state that "the component masses consistent with the signal lie roughly along a line of constant chirp mass" and label chirp mass on their confidence regions, but make no quantitative statement about posterior orientation. Ohme et al. [5] apply principal component analysis to the post-Newtonian phase coefficients rather than to (m_1, m_2) posterior samples, predict no per-event orientation, and validate against no catalog. The `simple-pe` metric of Fairhurst et al. [6] assumes the match "varies quadratically with the difference in parameters" about a peak, yielding an error ellipse — the same local, linear class of approximation as our tangent term, though computed in $(\mathcal{M}, \eta)$ rather than (m_1, m_2) coordinates — and those authors themselves note they "observe curved contours rather than perfect ellipses". To our knowledge no published work reconstructs the per-event (m_1, m_2) posterior orientation from the chirp-mass curve over the event's own q marginal, or validates such a reconstruction out-of-sample across catalogs;

that application is the contribution here.

4 The residual is a real systematic

The $\sim 1°$ residual is not sampling noise. Bootstrapping the released posterior samples — drawing one index set and recomputing *both* the measured axis and the curve inputs from it, so their correlated Monte Carlo errors cancel — gives a median resolution of $0.19°$ on the difference, against a median discrepancy of $1.03°$: a ratio of about six. The constant-$\mathcal{M}_c$ curve is an excellent but imperfect description of posterior orientation, and the gap is a property of the model, not of the sample count.

We are careful about what this bootstrap measures. It is the *Monte Carlo resolution* of the released sample set: it shrinks as more samples are released and encodes nothing about detector noise, calibration, waveform systematics or population variation. It is therefore not coverage, and we do not quote the fractions of events falling within one or two bootstrap widths as though they were.

The residual carries a signed component — the curve under-rotates relative to the measured axis by a median $0.78°$ — but this is not uniform across catalogs: it is significant in the training catalog ($p < 0.001$) and in O4a ($p = 0.019$), and *not* significant in the newest and largest catalog, O4b ($p = 0.38$). We report it per catalog for that reason.

The residual is strongly tracked by a measurable geometric quantity. The per-event self-consistency violation defined above correlates strongly with the per-event orientation residual: Spearman $\rho = 0.68$ ($p = 5e - 12$) at our primary projection grid, and ρ between 0.64 and 0.68 across a sweep of grid resolutions, so the result does not rest on that tuning choice. This is a correlation between two independently measured quantities rather than a fit, so it cannot be inflated by added model freedom. We say *tracks* rather than *explains*: both quantities derive from the same posterior and the same curve, so the correlation localises the residual within the geometry without establishing a causal mechanism, and the corresponding correction does not pass out-of-sample. Replacing the curve by those conditional means — one Hastie–Stuetzle iteration — reduces the residual in-sample, but we do not claim it as an improved reconstruction: the improvement is grid-sensitive (median 0.35–1.08° across the sweep), and in a cross-waveform-family test it improves in both directions but reaches $p < 0.05$ in only one ($1.14° \to 0.66°$, $p = 0.024$; $1.00° \to 0.69°$, $p = 0.064$).

A second mechanism is supported. The curve model has zero thickness, whereas a real posterior has finite width perpendicular to the curve — a median of about a third of its total spread. Learning that width profile on one waveform family and using it to reconstruct another family's axis improves on the zero-thickness curve in both directions ($1.14° \to 0.96°$, $p = 0.005$; $1.00° \to 0.74°$, $p = 0.028$), so finite thickness is a genuine out-of-sample contributor. The *arc-varying* part is not established: a constant thickness performs as well as the measured profile in one direction, and simple linear or quadratic tapers match or beat it in both. We therefore attribute no specific fraction of the residual to the measured taper.

5 Frames, coordinates, and the elongation gate

The measured object is a Euclidean principal-axis angle in fixed coordinates and is not reparameterization-invariant; we therefore state its dependence rather than assume it away. Repeating the primary score in *detector-frame* masses gives $0.31°$ against $1.03°$ in the source frame (paired over the same 81 events, $p = 4 \times 10^{-9}$), and in $(\log m_1, \log m_2)$ gives $0.77°$. The detector-frame gain is largely *mediated by elongation*: removing the redshift uncertainty that source-frame masses import makes

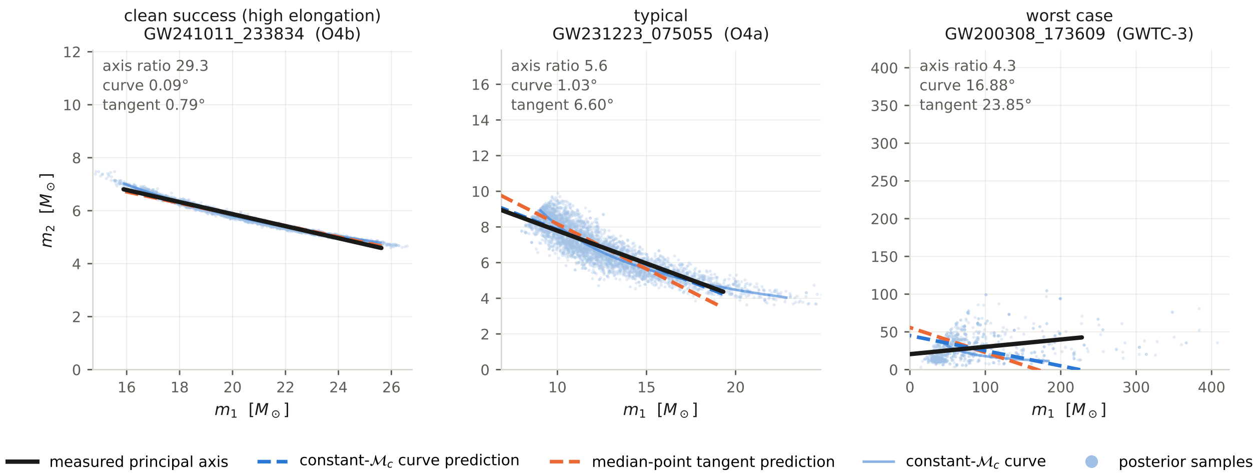

Figure 3: **Illustrative examples** of the posterior geometry and its reconstruction; this panel explains, it does not establish a mechanism. Each panel shows one event's (m_1, m_2) posterior samples, the constant-$\mathcal{M}_c$ curve through the event's median chirp mass, the measured principal axis, and the two predicted orientations. Axes use *equal aspect*: orientation angles are only visually truthful under equal aspect. Events are chosen by deterministic rules recorded in the figure sidecar, not selected by eye: *clean success (high elongation)* GW241011_233834 (O4b, axr 29.3, curve 0.09°, tangent 0.79°); *typical* GW231223_075055 (O4a, axr 5.6, curve 1.03°, tangent 6.60°); *worst case* GW200308_173609 (GWTC-3, axr 4.3, curve 16.88°, tangent 23.85°). The worst case is included by rule so the figure is not a gallery of successes. Across all elongated events the tangent in fact beats the curve for 16% of events, rising to 31% above axr = 8, where the curve's advantage shrinks. No thickness-adjusted axis is drawn: finite posterior thickness is supported out-of-sample but arc-varying thickness is *not* established (Sec. 4).

posteriors markedly more elongated (median axis ratio 9.9 versus 4.6), and at matched axis ratio the two frames perform comparably — the detector frame is in fact worse in the 3–5 band. We therefore say the difference is consistent with mediation by elongation, not that the detector frame suits the model better. The source frame remains the locked primary score; the detector-frame analysis is post-hoc. In $(\mathcal{M}_c, q)$ coordinates the construction is degenerate by design: a constant-$\mathcal{M}_c$ curve is a vertical line, so the reconstruction carries no per-event content there.

The restriction to elongated posteriors is physical, not a tuning knob. As the covariance approaches isotropy the principal axis ceases to exist, and the score degrades smoothly toward the random baseline: median error 14.5° for axr $\in [1, 1.5)$, 7.2°, 6.0°, 1.4° and 0.6° for successive bands. Nothing distinguishes the chosen threshold of 3: the curve error falls monotonically and smoothly across thresholds from 1.5 to 10.

Figure 3 shows what the numbers describe, for three events chosen by stated rules including the worst case in the sample.

5.1 Reproducibility and the cached extract

Every figure and every derived number in Sections 2.3–5 is computed from a single cached extract of the released posteriors, so that all downstream analyses share one deterministic input and can be regenerated in seconds without re-reading the parameter-estimation files. The extract draws a fixed number of samples per event *with replacement*; it is therefore a bootstrap resample of the released samples rather than a subsample of them, and quantities computed from it carry a corresponding resampling noise. The locked out-of-sample scores in Table 1 are quoted from the original full-sample analyses and are not affected. Where a cached number and a full-sample number both exist we give both.

6 Coherence as a diagnostic — and its limits

If the geometry is measurement optics, then *departures* from it are informative. We use two such departures diagnostically, and we are careful about what that licenses.

First, a physics-blind "outlier atlas": ranking the elongated O4b events by their curve residual and correlating against nine pre-declared axes (precession χ_p, aligned spin χ_{eff}, mass ratio, redshift, signal-to-noise ratio, chirp mass, total mass, secondary mass, and waveform-family disagreement) with Benjamini–Hochberg control and a partial correlation removing the axis-ratio confound, *no physical axis* survives. The only surviving correlate is waveform-model disagreement — a measurement systematic — with a sign indicating that the curve tracks the waveform *ensemble* consensus better than any single waveform tracks another.

Second, the geometric exponent fit of Section 7 shows an apparent 3σ offset that we do not interpret as physics.

What licenses that second conclusion is not coherence by itself. Coherence across analyses can diagnose shared modelling or systematic structure, because a systematic common to every event will appear in every event. But so will a real physical effect: a genuine departure from the leading-order inspiral phasing would also be coherent across a catalog, so "coherent, therefore systematic" is not a valid inference and we do not use it as one. What actually demoted the exponent offset was specific and checkable: the two catalogs disagree with each other by as much as the offset from the general-relativistic value, so the bootstrap statistical error understates the true spread; and the effect vanishes in the clean limit, the most-elongated half returning a value consistent with the reference. Those are the discriminants, not the coherence.

We treat these as cautionary diagnostics rather than as a method with a theorem behind it. The project history contains a case that makes the point in the other direction: a Bayesian ringdown analysis carried out during this work produced an apparently stable, well-behaved result that was retracted once its posterior was found to be prior-dominated — stability and coherence had looked like evidence and were not. That episode is recorded in the project's public history; it forms no part of the results here.

7 An exploratory geometric diagnostic — and why it is not a test of GR

The exponents of $\mathcal{M}_c$ are fixed by the 0PN quadrupole formula. Generalize to the one-parameter family

$$\mathcal{M}_p = (m_1 m_2)^p (m_1 + m_2)^{1-2p}, \qquad \text{GR: } p = 3/5 = 0.600, \tag{2}$$

and fit p to the posterior orientations: because orientation is scale-free ($m_1(q;p) \propto q^{-p}(1+q)^{2p-1}$), the fit isolates the exponent from the mass scale.

This is not a test of general relativity, and we do not present it as one. The posterior samples were themselves produced with general-relativistic waveform models; fitting an effective exponent to their geometry probes the internal consistency of those inferences, not the theory that generated them. No bound on modified inspiral phasing follows. We retain the exercise only because it is a sensitive diagnostic of the reconstruction's own imperfections. On the O4b elongated events alone we find $p = 0.616 \pm 0.011$; combining O4a and O4b, $p = 0.628 \pm 0.009$.

Taken naively this is a coherent 3.1σ departure from GR — and it is a false alarm. Two pre-committed checks demote it. (i) The two disjoint event catalogs disagree with each other by 0.031, as large as the offset from GR (0.028); this catalog-to-catalog spread is a systematic that the bootstrap statistical error (0.009) does not capture. Including it, $p = 0.628 \pm 0.009\,(\mathrm{stat}) \pm 0.016\,(\mathrm{syst})$, or 1.5σ from GR. (ii) The per-event exponent depends on elongation (Spearman $= -0.41$, coherent in both catalogs); the cleanest, most-elongated half yields $p = 0.606$, consistent with the GR value. The offset is therefore a finite-width imperfection of the leading-order geometric model, coherent across catalogs, and vanishing in the clean limit. It is a diagnostic of the model, not a measurement of gravity, and it belongs in an appendix rather than in the headline claims.

8 Context: the information anatomy of a detection

This section reports a separate, forward-model calculation. It is included as context for the measurement optics picture and does not support the reconstruction claim of Sections 2–5; no artifact in this paper links the two. The same Fisher structure that motivates the posterior orientation also fixes *when*, in frequency, each parameter's information arrives. Computing the cumulative Fisher information for the chirp mass (0PN), mass ratio (1PN) and spin (1.5PN) across all 190 real O4a+O4b events, the ordering $\mathcal{M}_c < \eta < \chi$ holds universally: the chirp mass is always determined earliest ($\sim$ 35 Hz), spin latest, and the frequency separation — the anatomy's "richness" — is a monotonic function of total mass (Spearman -1.00), widest for light, long-inspiral systems and compressed for heavy ones that merge immediately in band. This is a frequency-resolved, per-event map of what each detection measured and when. It is computed *from* the waveform model rather than measured from strain — the monotonic ordering is therefore a property of the model, not an empirical finding, and the Spearman value of -1.00 reflects that determinism rather than a statistical result. A fully data-driven version (re-analysing loud events in frequency slices and comparing empirical to GR-predicted information accumulation) is left to future work.

9 Context: the loudest event, GW250114

Context only. This event lies outside the elongated set used for the reconstruction, and the ringdown results quoted below are other groups' work, reported here for orientation. Nothing in this section is evidence for the claims of Sections 2–5. GW250114 — the loudest event to date (network SNR $\sim$ 80), a near-equal-mass binary black hole with remnant $M_f^{\mathrm{src}} \approx 62.7\,M_\odot$, $\chi_f \approx 0.68$ — is essentially a much louder counterpart of the first detection GW150914. Its remnant predicts Kerr quasi-normal-mode frequencies of 248.9 Hz ($\tau = 4.10$ ms) for the fundamental and 243.4 Hz ($\tau = 1.35$ ms) for the first overtone; a published ringdown analysis using an orthonormal-mode basis reports the first overtone at 99.9% significance — up from 82.5% in nonorthogonal analyses of the same event — and finds "no significant deviation" from the Kerr prediction [9]. We quote the orthonormal figure with that context because the significance is a property of the analysis basis, and we describe the Kerr result as a null, which is what is reported, rather than as a confirmation of the no-hair theorem. Being near-equal-mass, its posterior is nearly round and it is (by construction) outside the elongated set used for the reconstruction — an illustration that the loudest events are not always the most informative for a given geometric probe. We add that our own attempt at an independent ringdown analysis of this event was *retracted*: its posterior proved to be prior-dominated, the credible interval reproducing the prior rather than measuring the data. We therefore rely on the published analyses cited above and make no ringdown claim of our own.

10 Discussion

We have shown that the orientation of a compact-binary mass posterior can be reconstructed from two one-dimensional summaries of that posterior — its median chirp mass and its mass-ratio marginal — to about a degree on elongated events, with no coefficient calibrated on the validation catalogs, and that the reconstruction holds out-of-sample on two later, disjoint event catalogs. The result is not the trivial observation that a curve beats a line: substituting another event's marginal degrades it several-fold and the achieved error lies below the minimum of 300 permutations, while the reconstruction transfers between separately inferred waveform-family posteriors of the same event. Its residual is a genuine systematic about six times the Monte Carlo resolution of the released samples, to which finite posterior thickness is an out-of-sample contributor. We offer the result as a posterior-compression law and a systematics diagnostic.

Status of the claims. Because several of these statements were narrowed in the course of an internal submission-gate audit, we state their status plainly. The reconstruction, its baselines, its cross-waveform transfer, its frame dependence and its per-waveform-family stability are reproducible from committed artifacts. The residual analysis is a Monte Carlo resolution statement, not coverage. The companion precision law relating chirp-mass fractional width to signal-to-noise and mass is *exploratory and not established*: its pooled fit rejects the leading-order mass exponent, and the agreement that appears in a light-mass subset rests on a split chosen after seeing that rejection, so we do not claim it. The two out-of-sample scores are also not equivalent evidence: the O4b preregistration is timestamped in the public record before the data were opened, whereas the O4a preregistration entered our public repository in the same commit as its results and is attested only by a private history. Finally, O4a and O4b are *disjoint event catalogs*, not independent experiments: they share detectors, calibration, waveform families, priors and analysis conventions. A separate Bayesian ringdown analysis performed during this program was retracted after its posterior was found to be prior-dominated; it is recorded in the project history and forms no part of the results here.

Reproducibility. The figures in this paper are generated from committed machine-readable artifacts, and their captions are emitted from the same artifacts so that text and figure cannot drift apart. Raw parameter-estimation files are not distributed with the code; they are public and are listed with record numbers in the accompanying data-availability documentation. Analyses read a single cached extract of those files, regenerated in about five minutes by `src/e94_build_posterior_cache.py` once the releases are present; no analysis or figure module reads the raw files directly.

Honest limitations. (1) This is a posterior-compression result: the q marginal is taken from the posterior being reconstructed, so it is not a prediction from source medians or external physics. (2) It is validated on elongated events; round posteriors have no well-defined orientation and are excluded by design, and Baird et al. [3] note that constant chirp mass ceases to characterise the mass degeneracy so cleanly at higher masses, bounding where such a construction should be expected to work. (3) The $\sim 1^\circ$ residual is explained in the principal-curve sense — its size tracks the curve's failure of self-consistency — but the corresponding one-step correction is not established out-of-sample, and the arc-varying form of the thickness contribution is not established either. (4) The measured angle is coordinate- and frame-dependent, and we make no invariance claim. (5) The exponent fit of Sec. 7 is a diagnostic of GR-generated posteriors and cannot bound deviations from GR. (6) The information anatomy is model-predicted, not strain-extracted. (7) A first-principles derivation of the curve's width and curvature from the Fisher information remains open. (8) Waveform-model choice contributes a per-event orientation scatter of a few degrees, the dominant per-event systematic.

Data availability. All data are public. The analysis uses three releases: GWTC-2.1/3 PE (GWOSC/Zenodo), GWTC-4.0/O4a PE (Zenodo 16053484) and GWTC-5.0/O4b PE (Zenodo 20276106, 20348006); record numbers for all three are in the accompanying data-availability documentation. The authors' working copy reaches the GWTC-2.1/3 files through a symbolic link to a separate local repository; a fresh checkout must place those files directly, or GWTC-3 will be absent from every table without raising an error. Analysis code, locked pre-registrations, per-event results, and a reproducibility notebook accompany this work.

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